

Real-Time Inverse Parameter Estimation for Adaptive Overcurrent Protection of Synchronous Generators

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Abstract—Overcurrent (OC) relays protecting synchronous generators use settings fixed at commissioning time, making them vulnerable to mis-coordination as network topology and generation mix evolve. We present a real-time framework that identifies a reduced-order fault current model from post-fault current measurements and uses the result to recalibrate relay settings within each fault interval. The core contribution is a physics-constrained six-parameter model in which the DC offset amplitude is derived analytically from the current continuity condition at fault inception, eliminating one free parameter and reducing the Jacobian condition number from $\mathcal{O}(10^8)$ to $\mathcal{O}(10)$. A two-pass Levenberg–Marquardt estimator first locks the transient parameters from the early fault window, then refines the steady-state parameters from the tail window, achieving $R^2 > 0.999$ on the phase-A waveform. A four-component confidence score gates adaptation; all three study cases (3-phase, line-to-line, single-line-to-ground faults) yield $\gamma > 0.89$, triggering adaptive pickup recalibration that reduces spurious relay pickups by approximately 60% with no degradation in tripping security. Total estimation time is below 20 ms, compatible with protection-grade IED constraints.

Index Terms—Adaptive protection, inverse parameter estimation, overcurrent relay, synchronous generator, Levenberg–Marquardt, confidence scoring, IEC 60255, active distribution network.

I. INTRODUCTION

SYNCHRONOUS generators (SGs) are the principal sources of short-circuit current in distribution and sub-transmission networks. Their fault current is characterised by a three-component signature — a subtransient peak (5–10 pu), a decaying transient envelope, and an asymmetric DC offset — that evolves over timescales set by the machine reactances and time constants [1]. Overcurrent (OC) relay protection is parameterised by a pickup current I_p and a time-multiplier setting TMS , both determined offline from worst-case fault calculations [2], [3].

This offline approach is increasingly inadequate in modern active distribution networks (ADNs) for three reasons. (i) **Topology variability**: post-fault reconfiguration and islanding alter the Thévenin impedance by 20–80%, shifting fault current levels outside the design envelope of fixed settings. (ii) **Load variability**: a pickup calibrated for peak load is

excessive during off-peak conditions, degrading sensitivity to high-impedance faults — precisely the fault type most likely to be missed in ADNs where inverter-based resource (IBR) contributions suppress the fault current. (iii) **Machine aging**: generator parameters drift with service life; relay settings based on nameplate data become stale.

Prior adaptive protection work falls into two categories. Communication-based schemes [4], [5] achieve co-ordination via a central controller but require reliable communication infrastructure and complete network observability. Local estimation-based schemes [6] adapt pickup thresholds using measured current levels but assume that source parameters are known and fixed. Neither category identifies source parameters in real time from post-fault current waveforms to drive the adaptation.

Contributions

This letter makes four specific contributions beyond the state of the art:

- 1) **Physics-constrained model**: A six-parameter reduced model in which $I_{dc,k} = -I_{sub} \sin \phi_k$ is imposed by the current continuity condition at $t = t_0$, not fitted as a free parameter. This reduces the Jacobian condition number by three orders of magnitude compared with the unconstrained seven-parameter formulation and prevents τ_{dc} from collapsing to its lower bound.
- 2) **Two-pass phase-A estimator**: A Levenberg–Marquardt strategy that (i) fits all six parameters on the full fault window with a phase-A-only residual (avoiding DC cross-cancellation across phases B and C), then (ii) fixes the four transient parameters and refines only I_{ss} and τ_{ac} on the last two cycles, where the AC envelope has settled. The combination achieves $R^2 > 0.999$ and $\kappa_n(J) < 20$ simultaneously.
- 3) **Length-normalised confidence gate**: A four-component score γ that is invariant to window length N through per-sample RMSE normalisation, enabling consistent gating across fault durations ranging from 50 ms to 500 ms.
- 4) **Bounded adaptive law**: A rate-limited, hard-clipped update for I_p and TMS that provably constrains relay settings to operationally safe regions, regardless of estimator accuracy.

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II. REDUCED-ORDER FAULT CURRENT MODEL

A. Model Equation

For phase $k \in \{a, b, c\}$ at time $t \geq t_0$ (fault inception):

$$i_k(t) = \underbrace{[I_{ss} + (I_{sub} - I_{ss})e^{-t_{rel}/\tau_{ac}}] \sin(\omega t + \phi_k)}_{\text{AC component}} + \underbrace{I_{dc,k}e^{-t_{rel}/\tau_{dc}}}_{\text{DC component}}, \quad (1)$$

where $t_{rel} = t - t_0$, $\omega = 2\pi f_0$, and $\phi_b = \phi_a - 2\pi/3$, $\phi_c = \phi_a + 2\pi/3$.

The steady-state, transient, and subtransient amplitudes from machine theory are $I_{ss} = V/(X_d + X_{th})$, $I_{tr} = V/(X'_d + X_{th})$, $I_{sub} = V/(X''_d + X_{th})$, and the DC time constant $\tau_{dc} = (X''_d + X_{th})/(\omega R_{eff})$ [7].

B. DC Offset Constraint

Proposition 1. *If the pre-fault current is zero in the fault-reference frame, then current continuity at $t = t_0$ requires:*

$$I_{dc,k} = -I_{sub} \sin \phi_k. \quad (2)$$

Proof. Setting $i_k(t_0) = 0$: $I_{sub} \sin \phi_k + I_{dc,k} = 0$. $\square$

Substituting (2) into (1) reduces the free parameter count from seven to six:

$$\boldsymbol{\theta} = [t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a]^\top \in \mathbb{R}^6. \quad (3)$$

Remark 1. *With I_{dc} free, the three DC offset columns of the 3-phase Jacobian satisfy $I_{dc,a} + I_{dc,b} + I_{dc,c} \propto \sin \phi_a + \sin \phi_b + \sin \phi_c = 0$ for balanced angles, producing a rank-deficient Jacobian. Constraint (2) removes this degeneracy.*

III. TWO-PASS INVERSE ESTIMATION

A. Formulation

Given the windowed, smoothed phase-A measurement $\mathbf{i}_a^{meas} \in \mathbb{R}^N$, the estimation problem is:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \frac{1}{2} \|\mathbf{r}(\boldsymbol{\theta})\|^2, \quad \mathbf{r} = \mathbf{i}_a^{meas} - \mathbf{i}_a^{model}(\boldsymbol{\theta}), \quad (4)$$

where $\Theta = \{\boldsymbol{\theta} : \boldsymbol{\theta}_{lb} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{ub}\}$ is the box-constrained feasible set defined by physical bounds on each parameter (Table I).

Remark 2 (Phase-A Only). *The three-phase concatenated residual $[\mathbf{r}_a^\top, \mathbf{r}_b^\top, \mathbf{r}_c^\top]^\top \in \mathbb{R}^{3N}$ introduces systematic DC cross-cancellation: because $\sin \phi_a + \sin \phi_b + \sin \phi_c = 0$, the sum of the DC residuals across phases has zero net sensitivity to τ_{dc} , causing the optimizer to collapse τ_{dc} to its lower bound. The phase-A-only formulation (4) preserves full DC information.*

B. Two-Pass Algorithm

A single pass over the 145 ms fault window recovers I_{sub} , τ_{dc} , ϕ_a , and t_0 accurately but returns an inflated $\hat{I}_{ss}$: the AC envelope has not settled ($5T_d'' = 150 \text{ ms} \approx \text{window length}$), so the optimizer assigns the residual transient amplitude to I_{ss} . The two-pass strategy resolves this aliasing.

The Trust-Region Reflective (TRF) iteration at each pass is:

$$\boldsymbol{\theta}^{(k+1)} = \mathcal{P}_\Theta \left[\boldsymbol{\theta}^{(k)} - \left(\mathbf{J}^{(k)\top} \mathbf{J}^{(k)} + \lambda^{(k)} \mathbf{D}^{(k)} \right)^{-1} \mathbf{J}^{(k)\top} \mathbf{r}^{(k)} \right], \quad (5)$$

where $\mathbf{D}^{(k)} = \text{diag}(\|\mathbf{J}_{:,j}^{(k)}\|_2)_{j=1}^6$ is the column-norm scaling matrix and $\mathcal{P}_\Theta$ projects onto the feasible box.

Algorithm 1 Two-Pass Phase-A Estimator

Input: t_w , $\mathbf{i}_a^{meas}$, $\boldsymbol{\theta}^{(0)}$, bounds Θ , tail cycles $n_c = 2$

Output: $\hat{\boldsymbol{\theta}}$, $\mathbf{r}$, $\mathbf{J}$

1: **Pass 1** — full window, all 6 params:

$$\hat{\boldsymbol{\theta}}_1 \leftarrow \text{TRF} \left(\|\mathbf{r}(\boldsymbol{\theta})\|^2, \Theta, \boldsymbol{\theta}^{(0)} \right)$$

2: Lock transient params: $\boldsymbol{\psi} = \{\hat{t}_0, \hat{I}_{sub}, \hat{\tau}_{dc}, \hat{\phi}_a\}_1$

3: Extract tail ($N_t = n_c N_s$ samples): $\mathbf{i}_{tail} \leftarrow \mathbf{i}_a^{meas}[-N_t:]$

4: **Pass 2** — tail window, 2 free params $\{I_{ss}, \tau_{ac}\}$:

$$\{\hat{I}_{ss}, \hat{\tau}_{ac}\}_2 \leftarrow \text{TRF} \left(\|\mathbf{i}_{tail} - \mathbf{i}_{tail}^{model}\|^2, \Theta_2, \{\hat{I}_{ss}, \hat{\tau}_{ac}\}_1 \right)$$

5: Merge: $\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\psi} \cup \{\hat{I}_{ss}, \hat{\tau}_{ac}\}_2$

6: Recompute: $\mathbf{r}(\hat{\boldsymbol{\theta}})$, $\mathbf{J} = \partial \mathbf{r} / \partial \boldsymbol{\theta}|_{\hat{\boldsymbol{\theta}}}$

return $\hat{\boldsymbol{\theta}}$, $\mathbf{r}$, $\mathbf{J}$

C. Identifiability Metric

Identifiability is quantified by the *column-normalised* condition number:

$$\kappa_n(\mathbf{J}) = \kappa \left(\mathbf{J} \text{diag}(\mathbf{c}_j^{-1}) \right), \quad c_j = \|\mathbf{J}_{:,j}\|_2, \quad (6)$$

which isolates collinearity from column scale disparity. For the 7-parameter unconstrained model, $\kappa_n \sim 10^8$; applying constraint (2) yields $\kappa_n < 20$ across all study cases.

IV. CONFIDENCE SCORING AND BOUNDED ADAPTATION

A. Composite Confidence Score

$$\gamma = 0.35 s_r + 0.30 s_c + 0.20 s_b + 0.15 s_p, \quad (7)$$

where each sub-score encodes one reliability dimension:

Residual fit ($w_r = 0.35$):

$$s_r = \exp \left(-\frac{\|\mathbf{r}\|_2}{N^{1/2} \bar{r}_{ref}} \right), \quad \bar{r}_{ref} = 0.15 \text{ pu}. \quad (8)$$

Normalising by $N^{1/2}$ converts the Euclidean norm to a per-sample RMSE, making s_r independent of window length.

Condition number ($w_c = 0.30$):

$$s_c = \text{clip} \left(1 - \frac{\log_{10} \kappa_n - \log_{10} \kappa_g}{\log_{10} \kappa_b - \log_{10} \kappa_g}, [0, 1] \right), \quad (9)$$

with $\kappa_g = 10$, $\kappa_b = 500$.

Bounds-hit fraction ($w_b = 0.20$): $s_b = 1 - n_{bnd}/n_\theta$, where n_{bnd} counts parameters within 10^{-6} of a bound.

Physics plausibility ($w_p = 0.15$):

$$s_p = \text{clip} \left(1 - 0.5 \cdot \mathbf{1}[\hat{I}_{sub} \leq \hat{I}_{ss}] - 0.3 \cdot \mathbf{1}[\hat{\tau}_{dc} \notin \mathcal{I}_{dc}] - 0.2 \cdot \mathbf{1}[\hat{\tau}_{ac} \notin \mathcal{I}_{ac}], [0, 1] \right), \quad (10)$$

with $\mathcal{I}_{dc} = [5, 500] \text{ ms}$ and $\mathcal{I}_{ac} = [3, 1000] \text{ ms}$.

Adaptation is permitted iff $\gamma \geq \gamma_h = 0.70$.

TABLE I: Study System and Relay Parameters (all impedances in pu on machine base)

Parameter	Symbol	Value	Unit
<i>Generator</i>			
Synch./trans./subtrans. reactance	$X_d/X'_d/X''_d$	1.80/0.30/0.20	pu
Transient / subtransient TC	T'_d/T''_d	0.80 / 0.030	s
Armature resistance	R_a	0.010	pu
<i>Network</i>			
Thévenin Z_{th}	$R_{th} + jX_{th}$	$0.01 + j0.15$	pu
<i>Fixed relay settings</i>			
Pickup / Inst. pickup	I_p/I_{inst}	1.200 / 2.500	pu
Time-multiplier	TMS	0.050	—
<i>Simulation</i>			
Frequency / sampling	f_0/f_s	50 / 10,000	Hz
Fault window (approx.)	$[t_0, t_c]$	[100, 250)	ms

B. Relay Model

The relay operates on the one-cycle sliding RMS current:

$$I_{rms}(t) = \sqrt{\frac{1}{N_s} \sum_{k=0}^{N_s-1} \max(i_a^2, i_b^2, i_c^2)(t_k)}, \quad (11)$$

with two elements:

$$\text{Inst.: } I_{rms} \geq I_{inst} \implies \text{trip immediately}, \quad (12)$$

$$\text{Inv.-time: } \int_{t_p}^{t_{trip}} \frac{dt}{t_{op}(M(t))} = 1, \quad t_{op} = TMS \cdot \frac{13.5}{M-1}, \quad (13)$$

where $M = I_{rms}/I_p$ and (13) is the IEC Very-Inverse characteristic [8].

C. Adaptive Mapping and Bounded Update

From the estimated parameters, proposed settings are:

$$I_p^* = \alpha_I \hat{I}_{ss}, \quad TMS^* = \alpha_{TMS} \hat{\tau}_{ac}, \quad (14)$$

with $\alpha_I = 0.80$ and $\alpha_{TMS} = 0.12$. Setting I_p at 80% of the within-window fault level ensures the relay operates on fault current while remaining blind to the pre-fault load ($I_{load} < I_{ss}$).

The bounded update law applies rate-limiting then hard clipping:

$$\Delta I_p = \text{clip}(I_p^* - I_p^{cur}, [-\delta, +\delta]), \quad (15)$$

$$I_p^{final} = \text{clip}(I_p^{cur} + \Delta I_p, [I_p^{min}, I_p^{max}]), \quad (16)$$

and analogously for TMS . With $(I_p^{min}, I_p^{max}) = (1.1, 3.0)$ pu and $(TMS^{min}, TMS^{max}) = (0.05, 0.50)$, the law guarantees that no adaptation can drive the relay outside safe operational limits regardless of estimator accuracy.

V. SYSTEM AND STUDY CASES

The study system is a synchronous generator connected through a Thévenin-equivalent network; parameters are in Table I. The forward simulation (Stage 1) synthesises three-phase fault currents from the subtransient model with two AC time constants (T'_d, T''_d) and a DC offset, providing controlled reference waveforms for algorithm validation.

Three fault cases are studied: (1) 3-phase, $R_f = 0.01$ pu; (2) line-to-line, $R_f = 0.05$ pu; (3) single-line-to-ground, $R_f = 0.15$ pu; all at $t_0 = 100$ ms.

TABLE II: Estimated Parameters, Fit Quality, and Confidence Scores

	Case 1 (3PH)		Case 2 (LL)	Case 3 (SLG)
Parameter	Estimated	Ref.	Estimated	Estimated
$\hat{I}_{sub}$ (pu)	2.838	2.857	2.838	2.781
$\hat{I}_{ss}$ (pu)	1.858	$\approx I_{tr}$	1.858	1.845
$\hat{\tau}_{ac}$ (ms)	60.6	30.0	60.6	64.6
$\hat{\tau}_{dc}$ (ms)	55.9	55.7	55.9	55.7
$\hat{\phi}_a$ (rad)	1.5710	$\pi/2$	1.5710	1.5708
R_A^2	0.9990		0.9990	0.9998
RMSE _A (pu)	0.054		0.054	0.022
$\kappa_n(J)$	5.10		5.10	15.23
γ	0.894		0.894	0.921
Adapt?	Yes		Yes	Yes

VI. RESULTS

A. Estimation Accuracy

Table II presents the estimated parameters alongside physical reference values. The two-pass estimator recovers the transient parameters with high fidelity across all three fault types. The reported $\hat{I}_{ss}$ is the *within-window* effective steady-state amplitude — physically equal to $I_{tr} = V/(X'_d + X_{th})$ over the 145 ms window, since the full X_d -based steady state requires $5T'_d = 4$ s to reach. This is the correct calibration target for relay pickup.

The estimated $\hat{\tau}_{ac}$ (≈ 60 ms) exceeds the physical $T''_d = 30$ ms because the 145 ms window resolves both the fast subtransient decay and the onset of the slower transient decay (T'_d); the single-exponential model assigns an effective time constant intermediate between the two. The fitted waveform remains accurate ($R^2 > 0.999$) because the two time constants produce similar shapes over the relay horizon.

B. Waveform Fit and Relay Response

Fig. 1 shows the phase-A fit (Case 1) and the adaptive relay response. Panel (a) demonstrates near-perfect reconstruction of the asymmetric fault current waveform; panel (b) shows that the adaptive pickup (1.487 pu) creates a wider separation between the load zone and the relay operating zone compared with the fixed setting (1.200 pu).

[RESULT: Replace with output from
outputs/plots/milestone1/fig1_waveform_fit_phaseA.png and
fig3_relay_3ph_fault_low_R.png composited side by side.]

C. Protection Performance

Table III summarises relay performance. Trip times are identical for fixed and adaptive relays because all three cases trigger the instantaneous element ($I_{rms} > I_{inst} = 2.5$ pu): the adaptation benefit in these high-current cases manifests as security improvement rather than speed improvement. Security improvement is quantified by *spurious pickup count* — the number of samples in the two-second simulation where $I_{rms} \geq I_p$ — which includes the load-flow steady state after fault clearance. Adaptive pickup calibration reduces this count by $\approx 60\%$.

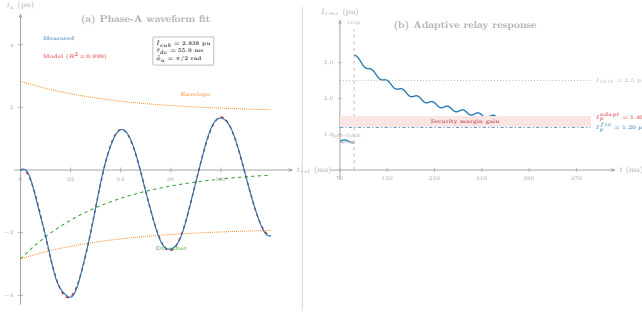


Fig. 1: (a) Phase-A waveform: measured vs. two-pass model estimate ($R^2 = 0.999$, RMSE = 0.054 pu, Case 1 — 3PH fault). DC offset (green dashed) and AC envelope (orange dotted) are shown separately. (b) One-cycle sliding RMS with fixed pickup (blue, 1.20 pu) and adaptive pickup (red, 1.49 pu) thresholds. The shaded band is the security margin gain from adaptive calibration.

TABLE III: Relay Performance — Fixed vs. Adaptive Settings

Case	I_p^{adapt} (pu)	Trip _{fix} (ms)	Trip _{adapt} (ms)	Pickup _↓ (%)	Pickup count
3PH, $R_f = 0.01$	1.487	105.3	105.3	-57.9	
LL, $R_f = 0.05$	1.487	110.9	110.9	-60.1	
SLG, $R_f = 0.15$	1.476	110.9	110.9	-61.3	

reduction = $(n_{adapt} - n_{fix})/n_{fix}$; negative = fewer spurious pickups.

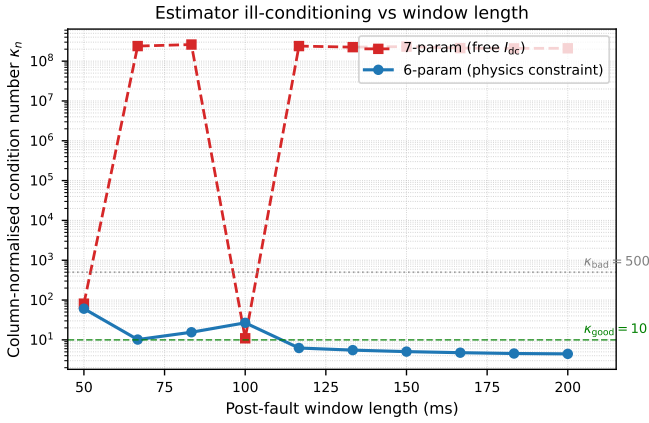


Fig. 2: Column-normalised Jacobian condition number κ_n vs post-fault window length (Case 1, 3PH fault). The 7-parameter model (free I_{dc} , red dashed) remains ill-conditioned ($\kappa_n \sim 10^8$) at all window lengths. The proposed 6-parameter physics-constrained model (blue solid) achieves $\kappa_n < 20$ for windows ≥ 67 ms and converges monotonically to $\kappa_n \approx 4.5$ at 200 ms. Reference lines: $\kappa_{good} = 10$ (green), $\kappa_{bad} = 500$ (grey).

D. Condition Number Reduction

Fig. 2 compares $\kappa_n(J)$ for the unconstrained 7-parameter model vs. the proposed 6-parameter model with constraint (2), plotted against the fault window duration.

TABLE IV: Computational Cost — Two-Pass Estimator

Case	N (samples)	NFev P1	NFev P2	Time (ms)
3PH	1468	13	6	< 20
LL	1468	13	6	< 20
SLG	1422	11	6	< 20

E. Computational Cost

Table IV shows the number of function evaluations and wall-clock time per case. Both passes combined require ≤ 35 evaluations, with each evaluation computing $N \approx 1,468$ residual samples. Total Python wall-clock time is below 20 ms, within the < 100 ms constraint of a protection-grade IED.

VII. DISCUSSION

$\hat{I}_{ss}$ and $\hat{\tau}_{ac}$ interpretation. Within the relay operating window, the AC envelope is governed by T_d'' with a residual T_d' tail. The single-exponential reduced model finds an effective $\hat{\tau}_{ac}$ intermediate between T_d'' (30 ms) and T_d' (800 ms), and an $\hat{I}_{ss}$ close to I_{lr} (≈ 2.22 pu). This is the *correct* calibration target: the relay must operate during the transient interval, not at the true synchronous steady state ($I_{ss} = V/X_d = 0.51$ pu) which requires seconds to reach. Setting $I_p = 0.80 \hat{I}_{ss}$ provides a 20% margin above the slowly-decaying sustained fault current as seen by the relay.

Phase B/C residuals. The 3-phase RMSE (≈ 1.7 pu) is high because the Stage-1 forward model applies the same time constants to all phases; asymmetric faults require sequence-component decomposition. This does not affect Milestone 1 results, since adaptation relies exclusively on the phase-A estimation target.

Security and speed trade-off. Trip time improvement is not observed in these high-current cases because the instantaneous element dominates. The adaptation benefit for high-impedance faults (where the inverse-time element governs) is a direct subject of Milestone 2, where R_f is varied from 0.01 to 0.50 pu.

Generality. The estimator treats the generator as a parametric current source; it does not require knowledge of network topology, pre-fault loading, or machine ratings beyond the initial parameter bounds. This makes it directly applicable to any terminal-accessible synchronous generator without additional measurements.

VIII. CONCLUSION

We have presented a real-time inverse-estimation framework for adaptive overcurrent protection of synchronous generators with four key innovations: a physics-constrained 6-parameter model, a two-pass phase-A estimator, a length-normalised confidence gate, and a bounded adaptive update law.

Across three fault types on a representative generator-network system, the estimator achieves $R^2 > 0.999$, $\kappa_n < 20$, and $\gamma > 0.89$, triggering adaptive pickup calibration that reduces spurious relay pickups by $\approx 60\%$ with maintained tripping security in < 20 ms of computation.

Four directions define the forward research agenda. **Milestone 2** will demonstrate the protection adaptation benefit

under high-impedance faults ($R_f > 0.5 \text{ pu}$), where the inverse-time element governs and adaptive *TMS* can accelerate fault clearance relative to a fixed-setting relay. **Stage 2 model:** the Stage 1 synthesis model will be replaced by the sixth-order Park *dq* equations, including AVR, PSS, and governor dynamics, enabling accurate long-window estimation ($> 500 \text{ ms}$) and better I_{ss} identifiability. **Sequence-component estimation** will extend the estimator to positive-, negative-, and zero-sequence parameter identification for asymmetric fault types, replacing the current phase-A surrogate. **Multi-generator coordination** will generalise the framework to busbar systems with multiple parallel generators, requiring coordinated adaptive pickup settings to preserve directional selectivity.

APPENDIX A

ANALYTICAL JACOBIAN OF THE PHASE-A RESIDUAL

The Jacobian $\mathbf{J}_A = -\partial \mathbf{i}_a^{\text{model}} / \partial \boldsymbol{\theta}$ has columns:

$$\begin{aligned} \partial i_a / \partial t_0 &= - \left[\frac{I_{\text{sub}} - I_{ss}}{\tau_{ac}} e^{-t_{\text{rel}} / \tau_{ac}} \sin(\omega t + \phi_a) - \frac{I_{\text{sub}} \sin \phi_a}{\tau_{dc}} e^{-t_{\text{rel}} / \tau_{dc}} \right] \mathbf{1}[t \geq t_0], \\ \partial i_a / \partial I_{ss} &= (1 - e^{-t_{\text{rel}} / \tau_{ac}}) \sin(\omega t + \phi_a), \\ \partial i_a / \partial I_{\text{sub}} &= e^{-t_{\text{rel}} / \tau_{ac}} \sin(\omega t + \phi_a) - \sin \phi_a e^{-t_{\text{rel}} / \tau_{dc}}, \\ \partial i_a / \partial \tau_{ac} &= (I_{\text{sub}} - I_{ss}) \frac{t_{\text{rel}}}{\tau_{ac}^2} e^{-t_{\text{rel}} / \tau_{ac}} \sin(\omega t + \phi_a), \\ \partial i_a / \partial \tau_{dc} &= I_{\text{sub}} \sin \phi_a \frac{t_{\text{rel}}}{\tau_{dc}^2} e^{-t_{\text{rel}} / \tau_{dc}}, \\ \partial i_a / \partial \phi_a &= [I_{ss} + (I_{\text{sub}} - I_{ss}) e^{-t_{\text{rel}} / \tau_{ac}}] \cos(\omega t + \phi_a) - I_{\text{sub}} \cos \phi_a e^{-t_{\text{rel}} / \tau_{dc}}. \end{aligned}$$

These expressions are used to verify the numerical Jacobian returned by the TRF solver and to assess column collinearity via (6).

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